# TP 1 : Base de données

**Modèle relationnel**

*student*(*idstudent*, *firstname*, *lastname*, *field*, *average*, *absence*, *city*)

| *idstudent* | *firstname* | *lastname* | *field* | *average* | *absence* | *city* |
|---|---|---|---|---|---|---|
| 1 | Ali | Hassan | Mathematics | 17.5 | 2 | Rabat |
| 2 | Sarah | Mohammad | Economics | 18.3 | 0 | Rabat |
| 3 | Omar | Yussuf | Mathematics | 16.0 | 0 | Kénitra |
| 4 | Laila | Ahmad | Economics | 14.4 | 0 | Rabat |
| 5 | Khalid | Ali | Mathematics | 19.2 | 0 | Salé |
| 6 | Fatima | Salem | Economics | 15.7 | 10 | Kénitra |
| 7 | Noor | Hussein | Mathematics | 18.1 | 0 | Salé |
| 8 | Ibrahim | Zaid | Economics | 19.8 | 0 | Salé |
| 9 | Aya | Nasser | Mathematics | 14.5 | 2 | Salé |
| 10 | Huda | Fahim | Economics | 13.0 | 3 | Rabat |
| 11 | Salma | Jamal | Mathematics | 17.3 | 0 | Salé |
| 12 | Yassin | Othman | Economics | 12.4 | 1 | Rabat |
| 13 | Amina | Sami | Mathematics | 19.2 | 1 | Rabat |
| 14 | Hassan | Rami | Economics | 18.7 | 0 | Rabat |
| 15 | Tariq | Bassam | Mathematics | 15.9 | 10 | Kénitra |
| 16 | Lina | Aref | Economics | 14.2 | 0 | Rabat |
| 17 | Zainab | Hadi | Mathematics | 17.5 | 5 | Rabat |
| 18 | Rashid | Kamal | Economics | 19.1 | 0 | Kénitra |
| 19 | Aisha | Fadel | Mathematics | 12.6 | 0 | Rabat |
| 20 | Noor | Mahmood | Economics | 16.7 | 2 | Rabat |

*Table 1 – Extrait de la relation student*

**1.** Afficher :

- 1.1 La liste des étudiants.
- 1.2 Les villes des étudiants du centre.
- 1.3 Les étudiants qui n'étaient jamais absents.
- 1.4 Les prénoms et les noms des étudiants qui habitent à *Rabat*.
- 1.5 Les moyennes des étudiants qui n'habitent pas à *Rabat*.
- 1.6 La liste des étudiants de la filière *Mathematics* triée selon la moyenne en ordre décroissant.
- 1.7 Les trois premiers étudiants de la filière *Economics*.

**2.** Afficher :

- 2.1 La note moyenne des étudiants du centre.
- 2.2 La note moyenne de chaque filière.
- 2.3 La plus grande et la plus petite moyenne de la filière *Mathematics*.
- 2.4 La somme des absences en *Economics*.
- 2.5 Le nombre des étudiants du centre.
- 2.6 Le nombre des étudiants par ville.
- 2.7 Les villes dont le nombre des étudiants dépasse 10.

# Corrigé

## Question 1 : requêtes de sélection

### Question 1.1

```sql
SELECT * FROM student;
```

### Question 1.2

```sql
SELECT DISTINCT city FROM student;
```

### Question 1.3

```sql
SELECT * FROM student WHERE absence = 0;
```

### Question 1.4

```sql
SELECT firstname, lastname FROM student WHERE city = 'Rabat';
```

### Question 1.5

```sql
SELECT average FROM student WHERE city != 'Rabat';
```

### Question 1.6

```sql
SELECT * FROM student
WHERE field = 'Mathematics'
ORDER BY average DESC;
```

### Question 1.7

```sql
SELECT * FROM student
WHERE field = 'Economics'
ORDER BY average DESC
LIMIT 3;
```

## Question 2 : fonctions d'agrégation et regroupement

### Question 2.1

```sql
SELECT AVG(average) AS Moyenne FROM student;
```

### Question 2.2

```sql
SELECT field, AVG(average) AS Moyenne
FROM student
GROUP BY field;
```

### Question 2.3

```sql
SELECT MIN(average) AS Note_min, MAX(average) AS Note_max
FROM student
WHERE field = 'Mathematics';
```

### Question 2.4

```sql
SELECT SUM(absence) AS Total
FROM student
WHERE field = 'Economics';
```

### Question 2.5

```sql
SELECT COUNT(idstudent) AS Total FROM student;
```

### Question 2.6

```sql
SELECT city, COUNT(idstudent) AS nb_etudiants
FROM student
GROUP BY city;
```

### Question 2.7

```sql
SELECT city, COUNT(idstudent) AS nb_students
FROM student
GROUP BY city
HAVING nb_students > 10;
```